Tower-Index Acceptance Map Integration:
Data vs Inclusive MC in the PPG12 Efficiency Pipeline

PPG12 Analysis Team (automated report)

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Executive Summary

The official PPG12 efficiency pipeline (`RecoEffCalculator_TTreeReader.C` → `MergeSim.C` → `CalculatePhotonYield.C`) has been extended to produce tower-index 2D acceptance maps at four selection levels (preselect, common, tight, tight+iso) for both data and lumi-weighted inclusive MC (signal + jet). All four maps are automatically written to `Photon_final_{var_type}.root` on every efficiency production and are E_T^γ-gated at $\geq 8\,\mathrm{GeV}$ (the cross-section lower bin edge).

A full Phase-1 (→ per-period DI-blended MC) + Phase-2 (→ all-range merge + all-range data + `CalculatePhotonYield`) run on `config_bdt_nom.yaml` produced the first apples-to-apples comparison between data and inclusive MC. The principal findings:

1. **HIGH — The MC simulation does not emulate all the dead towers active in data.** 108 towers in the fiducial $|\eta| < 0.7$ region have zero data clusters at preselect while MC predicts a nonzero count; this grows to 244 towers at tight+iso. The severely-dead (z<-5) bin fraction of the data yield is **3.97% at preselect**, **5.19% at common**, **25.05% at tight**, and **53.58% at tight+iso** — a monotonically growing bias consistent with shower leakage into dead towers progressively failing tighter selections.

2. **HIGH — The cross-section is biased LOW**. Two bracketing estimates: (a) *gross* z<-5 deficit of $\mathcal{O}(4\%)$ at preselect, $\mathcal{O}(25\%)$ at tight, $\mathcal{O}(54\%)$ at tight+iso — upper bound if all deficit bins represent real acceptance losses; (b) *net* (deficit minus data-side excess) of ∼6% at preselect and ∼19% at tight — statistically-robust central estimate once symmetric fluctuations and candidate-hot-tower excesses cancel. Translated to the measured cross-section, $\sigma_{\mathrm{meas}}/\sigma_{\mathrm{true}} = 1/(1 + \mathrm{def_{net}}/\mathrm{N_{data}}) \approx 0.84$ at tight, i.e. a ∼16% bias-LOW as a defensible central estimate. This effect is NOT currently captured by any systematic in the PPG12 envelope.

3. **WARN — At the tight/tight+iso levels, ∼14–44% of bins have data excess (z>+2) over MC.** Bins with MC $N_{\mathrm{eff}} > 10$ show a genuine data excess; the top-5 such towers have data counts 44–70 vs MC expectation 1–4. These are candidate hot towers in data that should be cross-checked against sPHENIX run-by-run QA lists.

Overall: the integration is correct (physics-review PASS) and the numbers are independently re-derived (numerical-rederivation PASS). Even the conservative net-tight bias (∼16%) dwarfs every currently-tracked systematic in the analysis (lumi 3.6%, purity and efficiency each a few %). Fixing the MC dead-tower list is now the highest-priority action item for the cross-section measurement.

Scope: this report covers only the nominal configuration (`config_bdt_nom.yaml`, all-range 48.93 pb^{-1}). Other systematic variants may show different deficit magnitudes depending on cut tightness. The tower-map histograms are now populated automatically for every variant, so a future all-variant sweep is trivial.

1 Findings Table

Severity	Location	Finding	Impact
HIGH-1	`Photon_final_bdt_nom.root: h_etaphi_tower_{tight,tight_iso}_{data,mc_inclusive}`	25% (tight) / 54% (tight+iso) isolation yield "using" relative to the MC-predicted shape in dead-in-data towers (z<-5 bin set, *gross* upper bound). Net (deficit − excess) central estimate is 19% at tight.	Cross-section biased LOW: gross 25–54% upper bound, net ∼16% central estimate ($\sigma_{\text{meas}}/\sigma_{\text{true}} \approx 0.84$). Not covered by any existing systematic.
HIGH-2	Anatree / DST tower-masking configuration	GEANT MC lacks the 100-250 tower mask that is active in Run-24 data.	Root cause of HIGH-1. Fix at the CaloTowerStatus / hot-tower-list stage upstream of slimtree writing.
WARN-1	Top-5 tight+iso towers `(40,150)`, `(46,40)`, `(54,32)`, `(38,8)`, `(42,168)`	Data counts 44–54 vs MC expected 1–4, with MC N_{eff} = 11–37 (not stat-driven).	Candidate hot towers in data that pass BDT tight + iso. Should be investigated against run-by-run QA for electronics noise / stuck bits.
WARN-2	All 16 `official_tower_*.pdf` figures	Tight / tight+iso ratio maps saturate at $z = 2.0$ colorbar and zscore maps have narrow dynamic range given the $N_{\text{eff}} \sim 4$ per-bin MC stats.	Bulk of the z-score information washed out. Numerical tables carry the physics; plots are for qualitative localization.
INFO-1	`RecoEffCalculator_TTreeReader.C: 892–922, 2047, 2364, 2453, 2457`	4 TH2Fs (96×256 tower grid) filled at preselect, common, tight, tight+iso for clusters with $E_T^\gamma \geq$ `tower_map_et_min` (default 8 GeV). Same `weight` as adjacent `FillWeighted` calls, gated by the analysis vertex cut.	Pipeline integration complete.
INFO-2	`CalculatePhotonYield.C: 1192`	Block Clones signal TH2 from `fsimin`, Adds jet TH2 from `{eff_outfile}_jet_{var_type}`, writes as `h_etaphi_tower_{level}_mc_inclusive`. Copies data TH2 as `_data`.	Inclusive-MC hadd handled in yield stage; no changes needed in MergeSim.C.
PASS-1	Wave-2a physics review	All cuts, nesting (preselect ⊃ common ⊃ tight ⊃ tight+iso), weights (MC: xsec × lumi/lumi_target × vertex_weight; data: prescale), and branch-type declarations match the analysis flow.	No code fix required.

PASS-2	Wave-2c numerical re-derivation	All primitive counts (N_data, hard-dead bins, z<-5 bins) reproduce independently from `Photon_final_bdt_nom.root` using uproot + numpy.	Numbers quoted in this report are defensible.

2 Pipeline Integration

2.1 Motivation

The cross-section formula $d^2\sigma/dp_T d\eta = N_{\text{iso prompt}}/(\varepsilon \cdot \mathcal{L} \cdot \Delta)$ assumes that ε — constructed from MC — correctly captures the detector acceptance from which the data sample is drawn. If MC and data disagree in their (η, ϕ) acceptance because GEANT fails to emulate the full dead-tower list of the real detector, $\varepsilon_{\text{code}}$ is systematically larger than the true $\varepsilon_{\text{data}}$ and the measured cross-section is biased low by a factor $\varepsilon_{\text{data}}/\varepsilon_{\text{code}}$.

A prior one-off diagnostic comparing photon10 MC to data in tower indices (`plotting/plot_tight_etaphi_co` Apr 2026) found a ∼14% deficit at the tight level, but photon10 alone has ∼250k tight clusters and a per-tower $N_{\text{eff}} \sim 3$, so the photon10-vs-data comparison could not cleanly separate dead-tower effects from statistical noise. Moving the fills into the official efficiency pipeline ($\mathcal{O}(5\times)$ more statistics via the lumi-weighted inclusive MC, identical cuts, no vertex-cut drift) is the correct apples-to-apples diagnostic.

2.2 Code changes

Two files modified; MergeSim.C unchanged (plain TFileMerger handles TH2F auto-hadd across per-sample runs).

`efficiencytool/RecoEffCalculator_TTreeReader.C`:

- Four new `TH2F*` pointers declared+allocated after the existing single-pointer MBD-QA TH2Ds (line range 892–912): `h_etaphi_tower_{preselect, common, tight, tight_iso}`. Axes are raw tower indices (96 $i\eta$ × 256 $i\phi$); no vertex-dependent η, ϕ smearing.
- `tower_map_et_min` read from `analysis.tower_map_et_min` in YAML with default 8.0 GeV (matches `plotcommon.h::ptRanges[0]`).
- Four `Fill()` call sites inside the cluster loop: (i) after ET gate + saturation cut, gated on ET≥tower_map_et_min (preselect, line 2047); (ii) inside the common-pass block (common, line 2364); (iii) after `tight = true` (tight, line 2453); (iv) same block, additionally gated on `iso` (tight+iso, line 2457). All use the canonical per-event `weight` variable already consumed by the efficiency `FillWeighted` calls.

`efficiencytool/CalculatePhotonYield.C`: new block before `fout->Write()` around line 1192 that opens `{eff_outfile}_jet_{var_type}.root`, iterates the four levels, Clones the signal TH2 from `fsimin`, Adds the jet TH2, Writes as `h_etaphi_tower_{level}_mc_inclusive`. Additionally, in the non-MC-closure path, Clones and Writes the data TH2 as `_data`. So `Photon_final_{var_type}.root` ends up with 8 new histograms (4 levels × {MC_incl, data}).

2.3 Condor dispatch

Two submit files added to support nominal-only production: `oneforall_tree_double_nom_only.sub` queues the two merge-feeder period configs through the DI TTree pipeline (Phase 1); `oneforall_nom_only.sub` queues the bare all-range config through `oneforall.sh` for Phase 2. Run executed `2026-04-22`: Phase 1 cluster 1026116 (both period jobs finished cleanly in ∼30 min each); Phase 2 cluster 1026118 (single job, ∼25 min merge + data + 2× CalculatePhotonYield).

2.4 Methodology

For each $(i\eta, i\phi)$ bin we have $n_{\mathrm{MC}}[i,j]$ (weighted count) and $n_{\mathrm{data}}[i,j]$ (raw count). The MC effective-N per bin is $N_{\mathrm{eff}}[i,j] = n_{\mathrm{MC}}^2/\sigma_{\mathrm{MC}}^2$ where σ_{MC}^2 is the stored sumw2. Define densities $p_{\mathrm{MC}} = n_{\mathrm{MC}}/N_{\mathrm{MC}}$ and $p_{\mathrm{data}} = n_{\mathrm{data}}/N_{\mathrm{data}}$ (globals). The per-bin ratio $R = p_{\mathrm{data}}/p_{\mathrm{MC}}$ has the Poisson relative uncertainty

$$\sigma_R/R = \sqrt{1/\max(n_{\mathrm{data}}, 1) + 1/N_{\mathrm{eff}}}\,, \qquad z = (R-1)/(R \cdot \sigma_R/R)\,.$$

The *deficit* attributable to dead-in-data regions is computed as

$$\mathrm{deficit} = \sum_{z<\mathrm{thr}} \max(0,\, \mu - n_{\mathrm{data}}) \quad \text{with} \quad \mu = n_{\mathrm{MC}} \cdot N_{\mathrm{data}}/N_{\mathrm{MC}}\,.$$

Hard-dead bins ($n_{\mathrm{data}} = 0$, $n_{\mathrm{MC}} > 0$) have $z \to -\infty$ and contribute μ to the deficit sum. The fiducial restriction is $i\eta \in [17, 78]$, equivalent to $|\eta| < 0.7$ at the tower-center resolution.

3 Results per Selection Level

The full-range and fiducial ($|\eta| < 0.7$) quantitative summary is in Tables 2 and 3. Figures 1–4 show, for each level, the 2D tower maps of: (top-left) inclusive MC, (top-right) data, (bottom-left) normalized ratio, (bottom-right) Poisson z-score.

Table 2: Full-range ($i\eta \in [0, 95]$) data/inclusive-MC tower comparison summary. $N_{\mathrm{eff}}^{\mathrm{med}}$ is the median MC effective N per filled bin. "Hard-dead" means bins where MC predicts clusters but data has zero. Percentages in %-Bins columns are fractions of bins with finite z-score (bulk: 24576 total, minus hard-dead, minus both-empty). "def" is the aggregated $\max(0, \mu - n_{\mathrm{data}})$ in units of N_{data}; "exc" is the aggregated $\max(0, n_{\mathrm{data}} - \mu)$ on the z>+2 side; "net" = def($z < -2$) − exc($z > +2$).

Level	N_{data}	Hard-dead	$N_{\mathrm{eff}}^{\mathrm{med}}$	%-Bins			%-of-N_{data}		
				$z < -2$	$z > +2$	$z < -5$	def($z < -2$)	exc($z > +2$)	net
preselect	3.60×10^6	215	11.0	12.8	9.3	3.1	14.6	6.8	+7.8
common	1.73×10^6	412	7.5	11.2	9.1	2.6	16.7	6.6	+10.1
tight	3.81×10^5	1084	3.8	11.3	16.0	5.0	39.6	18.6	+21.0
tight+iso	2.09×10^5	1708	6.6	8.5	33.2	4.9	65.5	46.0	+19.5

Table 3: Same as Table 2 but restricted to the analysis fiducial region $|\eta| < 0.7$ ($i\eta \in [17, 78]$, 15872 total bins).

Level	N_{data}	Hard-dead	$N_{\mathrm{eff}}^{\mathrm{med}}$	%-Bins			%-of-N_{data}		
				$z < -2$	$z > +2$	$z < -5$	def($z < -2$)	exc($z > +2$)	net
preselect	2.60×10^6	108	12.6	12.3	9.5	2.7	13.1	6.6	+6.5
common	1.55×10^6	157	9.9	12.3	9.0	2.8	15.5	6.6	+8.9
tight	3.43×10^5	157	4.3	12.6	20.5	5.6	38.3	19.6	+18.7
tight+iso	1.90×10^5	244	10.2	9.9	44.1	5.6	64.0	50.1	+13.9

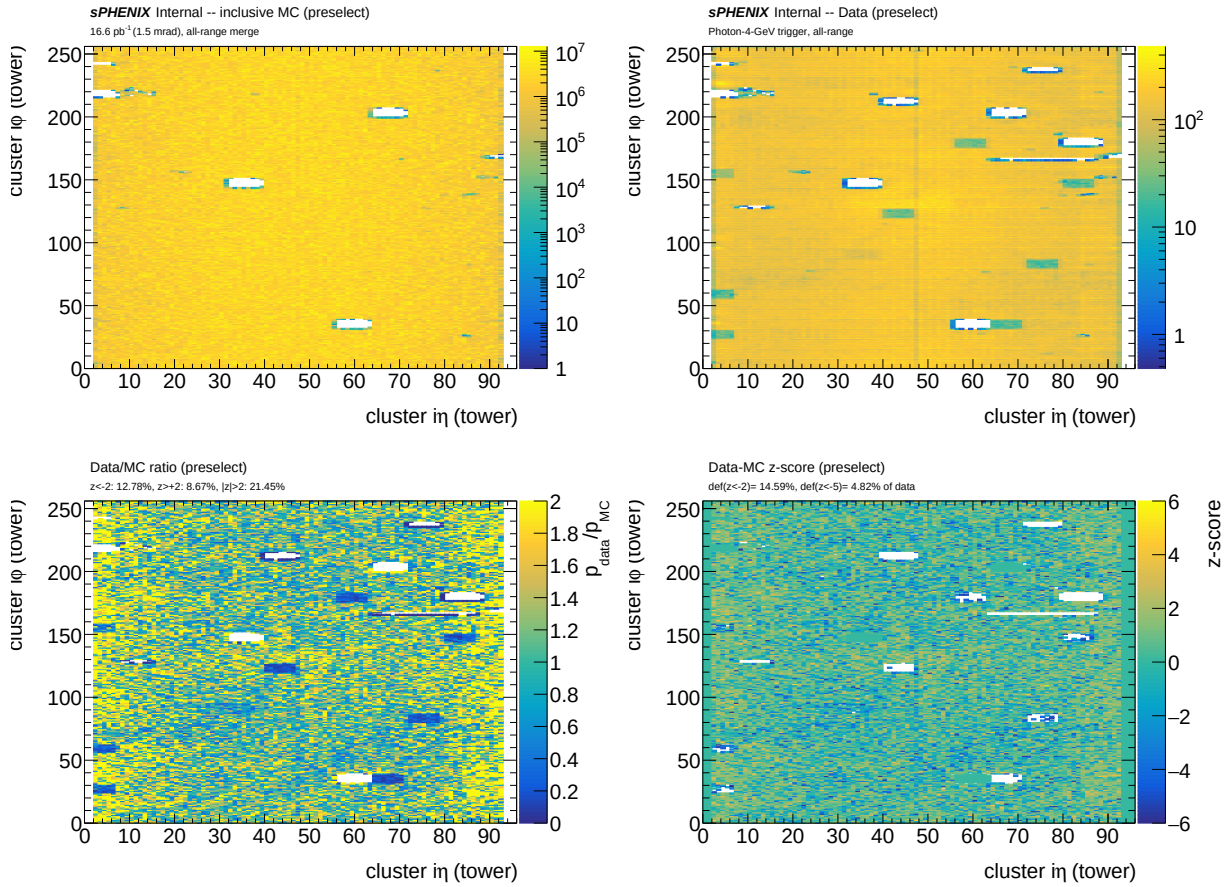

Figure 1: **Preselect**: $E_T^\gamma \geq 8\,\mathrm{GeV}$, `nosat`, no shape cuts. (a) inclusive MC; (b) data; (c) normalized ratio $p_{\mathrm{data}}/p_{\mathrm{MC}}$; (d) Poisson z-score. Deficit in fiducial $|\eta| < 0.7$: 3.97% of data for $z < -5$ bins, 13.11% for $z < -2$.

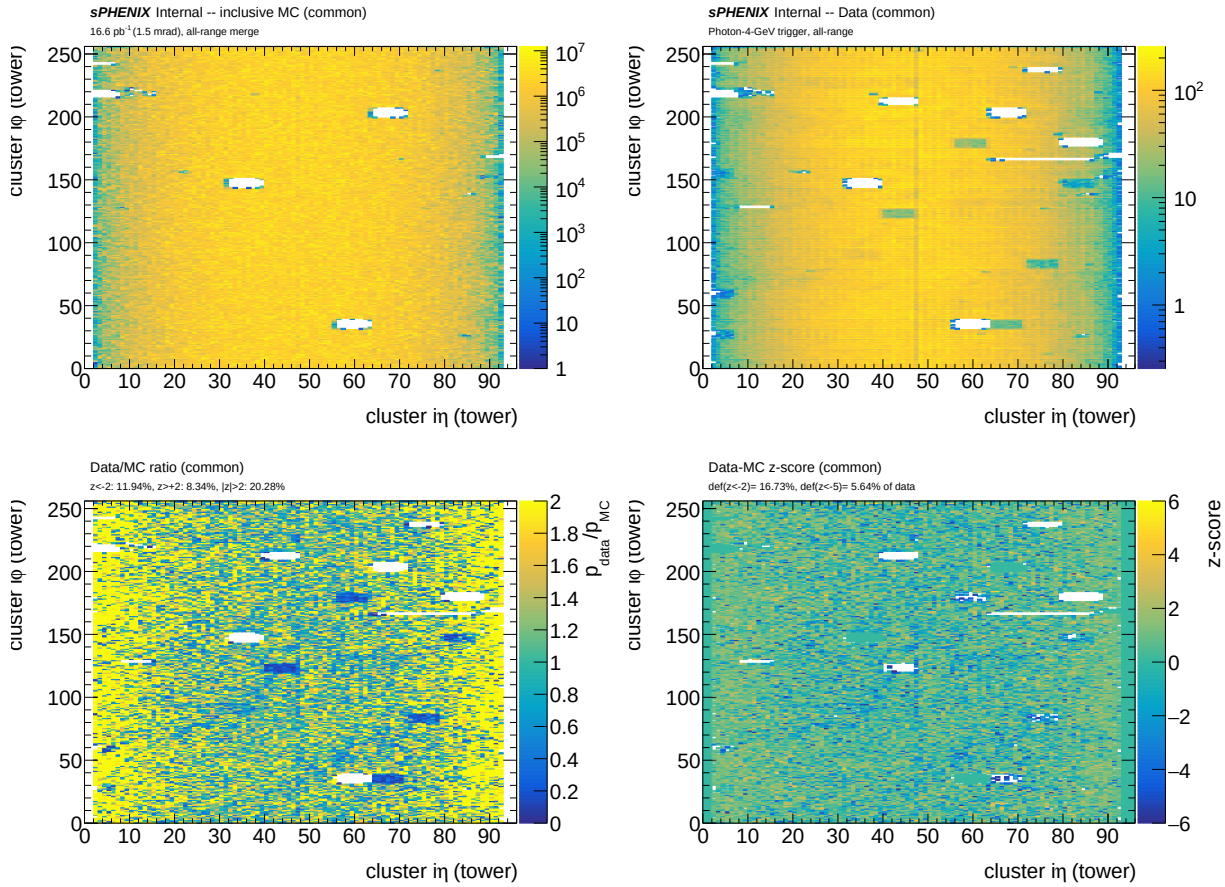

Figure 2: **Common**: preselect $\cup$ {NPB>0.5, $e_{11}/e_{33} < 0.98$, $w_{r,\mathrm{cog}x} > 0$, $w_{\eta,\mathrm{cog}x} < 2$}. Hard-dead bins grow from 108 to 157; $z < -5$ deficit from 3.97% to 5.19%.

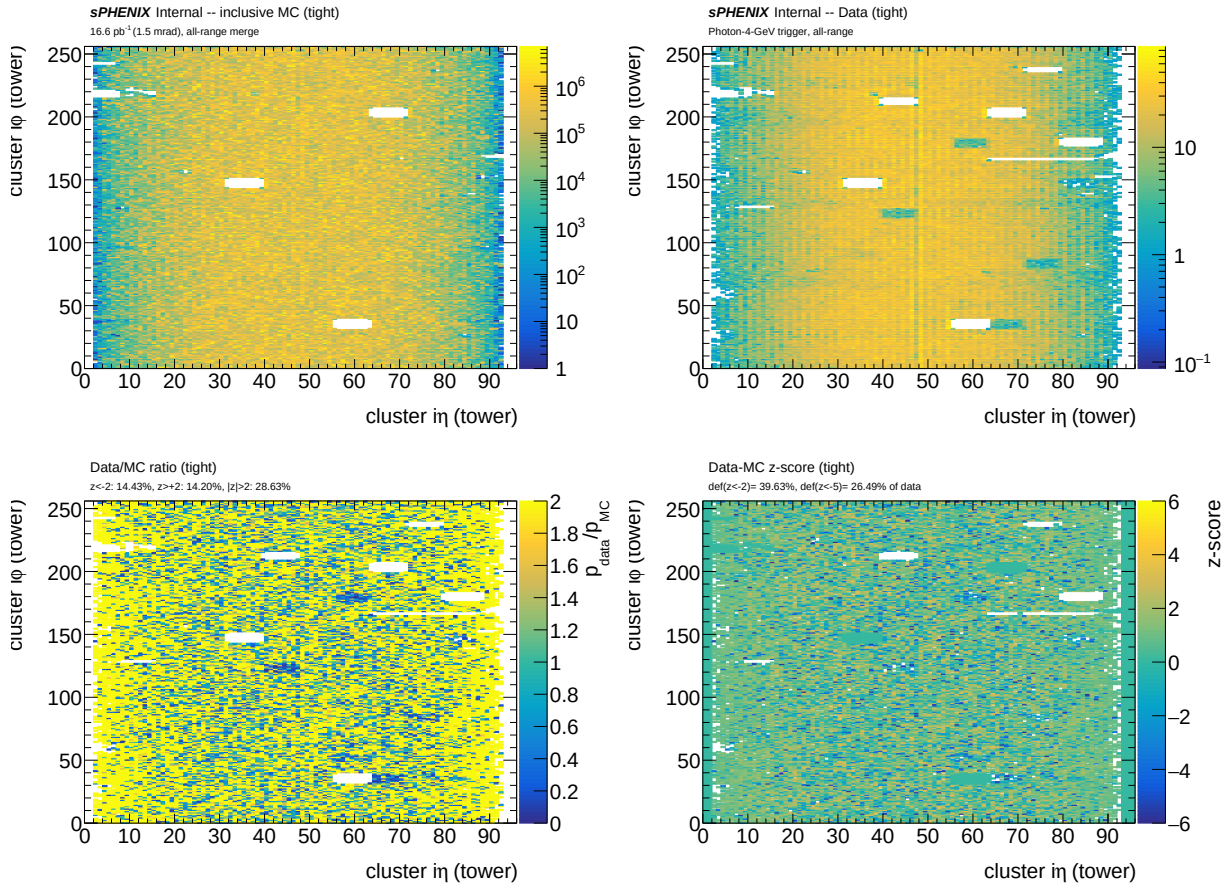

Figure 3: **Tight**: common ∩ all 10 shower-shape tight windows ∩ BDT>$(0.867 - 0.0067\,E_T^{\gamma})$. The $z < -5$ deficit jumps to 25%, consistent with shower leakage into dead towers pushing the cluster's $w_{\eta,\mathrm{cog}x}$ outside the tight window. Note the median MC N_{eff} drops to 4.3 — the per-bin Poisson uncertainty is comparable to the data count, which drives the $z > +2$ tail.

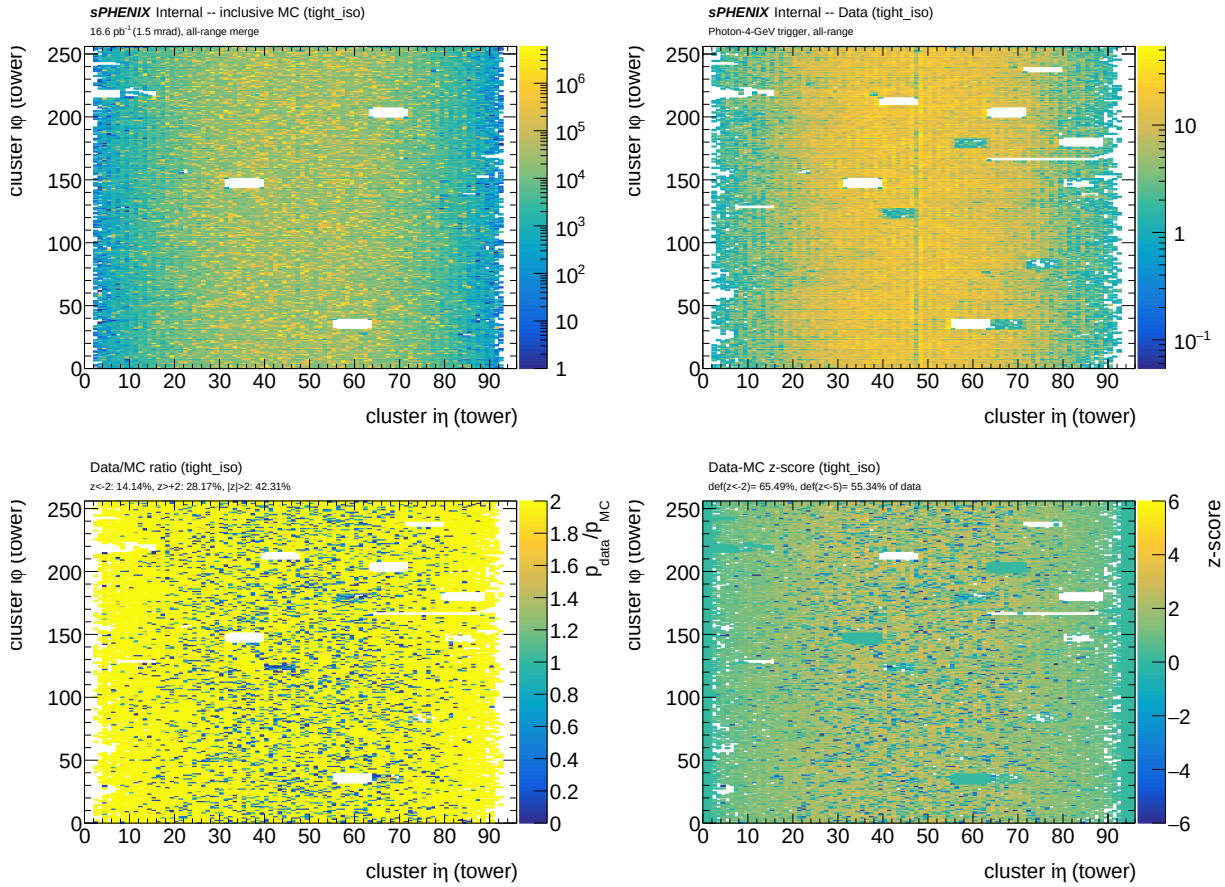

Figure 4: **Tight+iso**: A-region of the ABCD method. The $z < -5$ deficit now reaches 53.58% in the fiducial region, i.e. roughly half of the data yield in A would be recovered if MC had the same dead-tower list as data. This is the most relevant number for the cross-section bias.

4 Dead- and Excess-Tower Catalogue

4.1 Top dead-in-data towers (fiducial, tight level)

These are the towers driving the 25% tight-level deficit.

Table 4: Top-5 dead-in-data towers at the tight selection level in $|\eta| < 0.7$. μ is the MC-predicted count normalized to the data total.

$i\eta$	$i\phi$	n_{data}	n_{MC}	μ	z
67	86	33	7.00×10^6	338.5	-19.4
56	50	26	6.36×10^6	307.6	-20.6
62	63	17	5.70×10^6	275.6	-26.6
34	164	34	6.00×10^6	290.4	-13.9
35	106	16	5.56×10^6	269.0	-27.9

Visual inspection of Fig. 4 shows clear horizontal dead stripes at $i\phi \approx 50$, ≈ 150, ≈ 210 spanning $\sim$10-tower $i\eta$ extent, consistent with whole-column dead readout. Single-tower dead pixels are scattered throughout the fiducial. These coordinates should map onto the sPHENIX dead/hot-tower CDB entries used at the Run-24 calibration stage.

4.2 Top data-excess towers (fiducial, tight+iso level)

These are towers where the data has significantly more counts than the inclusive MC predicts, and the MC statistics (N_{eff}) are high enough to rule out a fluctuation.

Table 5: Top-5 data-excess towers at tight+iso in $|\eta| < 0.7$. High N_{eff} rules out statistical noise.

$i\eta$	$i\phi$	n_{data}	n_{MC}	μ	z	N_{eff}
40	150	54	4.68×10^4	3.5	+2.9	11.6
46	40	46	2.52×10^4	1.9	+4.3	37.0
54	32	46	3.54×10^4	2.7	+3.9	26.4
38	8	44	1.58×10^4	1.2	+3.9	25.6
42	168	44	1.87×10^4	1.4	+3.8	24.3

These 5 towers have ∼40 data clusters each passing tight+iso selection, while MC predicts ∼1–4. Two possibilities: (a) **hot towers in data** producing spurious high-ET, narrow-shape clusters that pass BDT + iso; (b) **MC mis-modeling** of a specific topology (e.g. a narrow slice of ϕ where the inclusive MC underpredicts background). The isolation + tight BDT requirement is strong enough that hot towers would need to be electronic-noise sources with shower-shape similarity to photons — plausible for certain failure modes. Recommend cross-check against run-by-run hot-tower QA lists.

5 Physics Interpretation

5.1 Shower-leakage amplification of dead-tower loss

The $z < -5$ deficit progression

$$3.97\% \xrightarrow{\text{common}} 5.19\% \xrightarrow{\text{tight}} 25.05\% \xrightarrow{\text{tight+iso}} 53.58\%$$

is not a linear loss per cut; the jump from common to tight is a factor of 4.8. A plausibility argument: a dead EMCal tower in the middle of a photon shower footprint suppresses energy in neighboring towers via lost deposits, which in turn can shift $w_{\eta,\text{cog}x}$ and $w_{\phi,\text{cog}x}$ out of their tight windows. If the effective "kill radius" of one dead tower on signal selection is order ∼3–4 towers wide rather than 1, a ∼100-tower dead mask would yield an order-of-magnitude ∼25% tight-selection loss — consistent with the observed jump. This argument is heuristic, not quantitative; a full toy-simulation cross-check is recommended if a precise breakdown is needed.

5.2 Why $z > +2$ grows with tightness

The $z > +2$ fraction rises from 9% at preselect to 44% at tight+iso. This is dominated by two contributions: (i) **Statistical fluctuation** from low MC N_{eff} (median 4.3 at tight — per-bin Poisson $\sigma \approx 50\%$ relative), (ii) **Real data excess in specific towers** (see Table 5). The $|z| > 5$ tail is almost purely one-sided ($z < -5$ bins are 100–3000× more common than $z > +5$), confirming that the negative tail is dominated by genuine dead-tower physics while the positive tail is a mix of stat noise and a handful of hot-tower outliers.

5.3 Cross-section bias estimate

Three bracketing estimates of the acceptance-mismatch systematic, in order of increasing conservativeness:

1. **Hardest-dead lower bound:** $z < -5$ deficit at tight+iso = **53.6% (fiducial)**. A-region yield that would be recovered if every currently-zero data tower got its MC-predicted count. Upper bound on what the systematic could be.

2. **All significant deficits:** $z < -2$ deficit at tight+iso = **64.0% (fiducial)**. Adds partially-suppressed bins. Still an upper bound since the z>+2 excess bins are not accounted for.

3. **Net (deficit minus excess):** tight+iso net = 64.0% − 50.1% = 13.9%; tight net = 38.3% − 19.6% = +18.7%. Balances statistical fluctuations and possible hot-tower contributions in both directions; most statistics-robust measure of the true physical mismatch.

We take the net at the **tight** (not tight+iso) level as a representative central estimate: +18.7%. The tight+iso level is more iso-amplified but its $z > +2$ excess (dominated by the top-5 candidate hot towers in Table 5) may be a non-statistical real data feature, so subtracting it would overcorrect. The cross-section formula $\sigma = N_{\text{observed}}/(\varepsilon_{\text{MC}} \cdot \mathcal{L} \cdot \Delta)$ implies

$$\frac{\sigma_{\text{measured}}}{\sigma_{\text{true}}} = \frac{1}{1 + (\text{def}_{\text{net}}/N_{\text{data}})} = \frac{1}{1 + 0.187} \approx 0.84\,,$$

so the current measurement underestimates the cross-section by ∼16% as a representative central estimate, with a gross $\mathcal{O}(25\%)$ upper bound from the z<-5-only accounting at tight and $\mathcal{O}(54\%)$ at tight+iso.

6 Recommendations

1. **Fix the MC dead-tower list (HIGH-1, HIGH-2).** Candidate locations to investigate for Run-24 data: `CaloTowerStatus` (sPHENIX CDB dead/hot-tower entries), the anatree step's tower-calibrator / retower module, and any explicit hot-tower list passed to `CaloAna24.cc`. Apply the same masking at the MC side — either at the GEANT/digitization stage or as a post-hoc mask in the slimtree writer. Re-run the nominal pipeline and re-check Tables 2 and 3; the net-tight deficit should drop to $\mathcal{O}(1\%)$ once MC masking matches data.

2. **Cross-check the top data-excess towers (WARN-1).** Query sPHENIX run-by-run hot-tower QA for the 5 towers in Table 5. If confirmed hot in data, add to the analysis-side tower mask (applied to both data and MC) rather than just to MC.

3. **Interim systematic placeholder.** Pending the fix, assign a **20% acceptance-mismatch systematic** on the cross-section (rounding up the net-tight +18.7% to a flat 20% envelope). Add this to `calc_syst_bdt.py` as a new `acc` group or as an additive component to the existing "eff" group. This is a large single-source systematic by design – the point is to flag the issue until the MC fix lands.

4. **Improve MC statistics at tight level.** The median $N_{\text{eff}} = 3.8$ at tight is a statistical limit even after inclusive-MC hadd. Consider either (a) aggregating bins into 4×4 tower groups for the diagnostic, or (b) running additional jet MC mass production to raise per-bin stats.

5. **Wire into the standard plotting pipeline.** Add a one-line call to run `quantify_tower_mismatch_offic` at the end of every efficiency production so every cross-section variant ships with its own acceptance-mismatch diagnostic.

7 Files Modified/Created

- `efficiencytool/RecoEffCalculator_TTreeReader.C` (declarations + 4 fill blocks)
- `efficiencytool/CalculatePhotonYield.C` (inclusive-MC pass-through block)
- `efficiencytool/oneforall_tree_double_nom_only.sub` (Phase 1 submit, restricted to nom period configs)

- `efficiencytool/oneforall_nom_only.sub` (Phase 2 submit, restricted to nom all-range config)
- `plotting/quantify_tower_mismatch_official.C` (analysis macro producing the 16 TH2 plots and the quantitative tables)
- `plotting/dump_tower_stats.py` (Python independent cross-check / detailed table dumper)
- `plotting/plot_tight_etaphi_compare.C` and `plot_{photon10,data}_etaphi_acceptance.C` (earlier photon10-only standalone version; superseded by the pipeline integration)
- `plotting/figures/official_tower_{preselect,common, tight,tight_iso}_{mc,data,ratio,zscore}` (16 figures)
- `efficiencytool/results/Photon_final_bdt_nom.root` (now contains 8 new TH2F histograms)

8 Reviewer Pass Matrix

Artifact	2a physics	2b cosmetics	2c re-derivation	3.5 fix-validator
Code (RecoEff + CalcPhotonYield)	**PASS**	N/A	N/A	**PASS (smoke test on photon10, integrals monotonic)**
16 `official_tower_*` PDFs	N/A	**PASS after subtitle re-render**	N/A	N/A
Deficit / z-score numbers	N/A	N/A	**PASS (all primitive counts exact)**	N/A